

## CHAPTER 8

# Sheaves of Boolean Germs: The Quantum Topological Case

### 8.1 Quantum Logical Event Algebras and Observables

The quantum case refers to a localization scheme on a global partial order of quantum physical events  $L$  taking place over a base category of reference contexts  $B$ . This base category consists of *Boolean algebras*  $B$  associated with the operational procedure of quantum measurements, the arrows between them being Boolean algebra homomorphisms. In this case, the reference contexts of the operational environment employed for observation are all the Boolean algebras  $B$ , whereas their morphisms are not inclusions but *all* Boolean homomorphisms. Equivalently, the arrows  $B \rightarrow L$  are considered as varying base reference frames of Boolean algebras over which a global partial order of quantum events  $L$  fibers.

Consider now the algebra  $L$  of *logical events or propositions* associated with the measurement of a quantum system. A *quantum logical event algebra*  $L$  in  $L$  is a partially ordered set of quantum logical events, endowed with a maximal element 1, and with an operation of orthocomplementation  $[-]^* : L \rightarrow L$ , which satisfy, for all  $l \in L$ , the following conditions:

$$[a] \quad l \leq 1$$

$$[b] \quad l^{**} = l$$

$$[c] \quad l \vee l^* = 1$$

$$[d] \quad l \leq l' \rightarrow l'^* < l^*$$

$$[e] \quad l \perp l' \rightarrow l \vee l' \in L$$

[f] for  $l, l' \in L$ ,  $l \leq l'$  implies that  $l$  and  $l'$  are compatible, where  $0 := 1^*$ ,  $l \perp l' := l \leq l'^*$ , and the operations of meet  $\wedge$  and join  $\vee$  are defined as usual. Recall that  $l, l' \in L$  are *compatible* if the sublattice generated by  $\{l, l^*, l', l'^*\}$  is a *Boolean algebra*, namely, if it is a *Boolean sublattice*. In order to have a well defined theory of observables over  $L$ , we also impose a  $\sigma$ -*completeness* condition, namely that the *join of countable families* of pairwise orthogonal events exists. It is important to notice here that a quantum logical event algebra  $L$  is *not* a Boolean algebra. It is technically called an *orthomodular  $\sigma$ -orthoposet*.

We can now construct a locally small cocomplete category, denoted by  $\mathbf{L}$ , which is called the category of quantum logical event algebras. The objects of  $\mathbf{L}$  are the quantum logical event algebras  $L$ , whereas the arrows of  $\mathbf{L}$  are quantum algebraic homomorphisms, defined as follows:

A *quantum logical algebraic homomorphism* in  $\mathbf{L}$  is a morphism  $H : K \rightarrow L$ , which satisfies, for all  $k \in K$ , the following conditions:

$$[a] \quad H(1) = 1$$

$$[b] \quad H(k^*) = [H(k)]^*$$

$$[c] \quad k \leq k' \rightarrow H(k) \leq H(k')$$

$$[d] \quad k \perp k' \rightarrow H(k \vee k') \leq H(k) \vee H(k')$$

A *quantum observable*  $\Xi$  is defined to be an algebraic morphism from the *Borel algebra* of the real line  $Bor(\mathbb{R})$  to the quantum event algebra  $L$ ,

$$\Xi : Bor(\mathbb{R}) \rightarrow L$$

such that:

$$[a] \quad \Xi(\emptyset) = 0, \Xi(\mathbb{R}) = 1$$

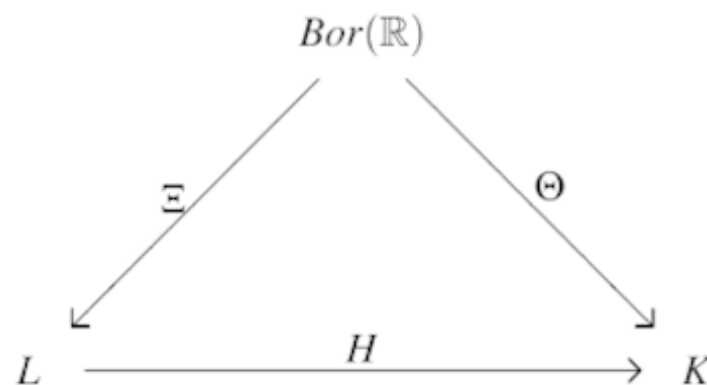
$$[b] \quad E \# F = \emptyset \Rightarrow \Xi(E) \perp \Xi(F), \text{ for } E, F \in Bor(\mathbb{R})$$

$$[c] \quad \Xi(\#_n E_n) = \vee_n \Xi(E_n), \text{ where } E_1, E_2, \dots$$

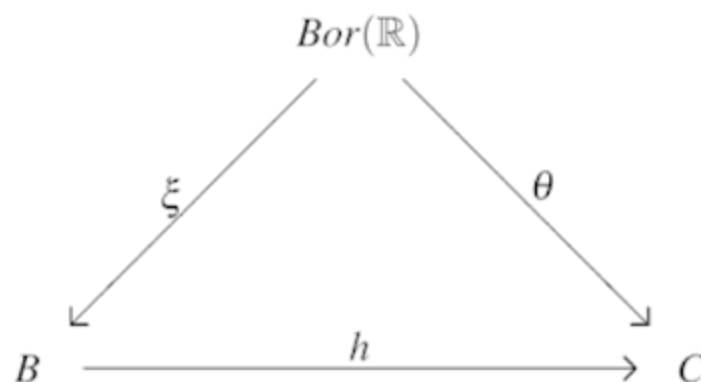
is a sequence of mutually disjoint Borel sets of the real line.

Addition and multiplication over  $R$  induce on the set of quantum observables the structure of a *partial commutative algebra* over  $R$ . In most of the cases the stronger assumption of a non-commutative algebra of quantum observables is adopted.

Thus, we may construct a category, denoted by  $O_q$ , which is called the *category of quantum observables*. Its objects are quantum observables  $\Xi: \text{Bor}(R) \rightarrow L$  and its arrows  $\Xi \rightarrow \Theta$  are commutative triangles, or equivalently the quantum algebraic homomorphisms  $H: L \rightarrow K$  in  $L$ , preserving by definition the join of countable families of pairwise orthogonal events, such that  $\Theta = H \# \Xi$  in the following diagram is again a quantum observable.



Correspondingly, we may construct a category, denoted by  $Ob$ , which is called the *category of Boolean observables*. Its objects are the Boolean observables  $\xi: \text{Bor}(R) \rightarrow B$  and its arrows are the Boolean algebraic homomorphisms  $h: B \rightarrow C$  in  $B$ , such that  $\theta = h \# \xi$  in the following diagram is again a Boolean observable.



If  $L$  is isomorphic with the orthocomplemented lattice of *orthogonal projections* on a complex Hilbert space, then from von Neumann's *spectral theorem* the quantum observables are in bijective correspondence with the hypermaximal Hermitian operators on the Hilbert space.<sup>1</sup> It is important to note, here, that the original quantum logical formulation of quantum theory depends in an essential way on the *identification of propositions with projection operators on a complex Hilbert space*. A non-

classical, *non-Boolean* logical structure is thereby induced which has its origins in quantum theory. More accurately, the Hilbert space quantum logic has been initially axiomatized as a complete, atomic, orthomodular lattice.<sup>2</sup> Equivalently, it could be cast as isomorphic to the *partial Boolean algebra of closed subspaces* of the Hilbert space associated with the quantum system, or alternatively the *partial Boolean algebra of projection operators* of the system.<sup>3</sup> By contrast, the propositional logic of classical mechanics is *Boolean logic*; that is, for the propositions of classical mechanics, the class of models over which validity and associated semantic notions are defined is the class of Boolean logic structures.

In this setting a fundamental result is provided by the Kochen-Specker theorem, according to which *there are no two-valued homomorphisms on a quantum logical event algebra  $L$  globally*. Thus, a quantum event algebra *cannot be embedded* into a Boolean algebra. Despite this admonition, however, and as stressed in chapter 2, in every concrete quantum measurement context, the set of logical events that can be actualized in this context *always* forms a Boolean event algebra. This fact motivates the assertion that *a Boolean algebra within the lattice of quantum logical events plays the topological role of a local logical reference frame* relative to which a measurement result is always coordinatized.

Likewise, together with a logical event structure, there always exists a corresponding *probabilistic structure*, defined by means of convex sets of measures on that logic. In this sense, the probabilistic structure of a classical system is described by convex sets of probability measures on the Boolean algebra of events of this system; whereas the probabilistic structure of a quantum system is described by convex sets of probability measures on the quantum logical event structure of that system.<sup>4</sup> More accurately, in the case of quantum systems, if the quantum event algebra is denoted by  $L$ , each *quantum probability measure*, or *quantum probabilistic state*, is defined by a mapping:

$$p: L \rightarrow [0,1]$$

such that the following conditions are satisfied:  $p(1) = 1$  and  $p(x \vee y) = p(x) + p(y)$ , if  $x \perp y$ , where  $x, y \in L$ . In the Hilbert space formulation of quantum theory,  $L$  denotes the Hilbert space quantum event algebra, whereas a quantum state is defined by the Hilbert space inner product:

$$\langle \varphi, Px\varphi \rangle$$

where  $x \in L$ ,  $\varphi$  is a normalized vector in the Hilbert space, and  $Px$  is the orthogonal projection operator corresponding to  $x \in L$ .

## 8.2 The Topological Significance of Boolean Contexts

The set-theoretic axiomatizations of quantum event algebras *hides the intrinsic topological significance* of Boolean localizing contexts in the formation of these structures. On the other hand, the operational procedures followed in typical quantum measurement situations are based explicitly in the preparation and employment of appropriate *Boolean contexts*. That is to say, even if according to the Kochen-Specker theorem there *does not exist a global two-valued truth-functional* assignment pertaining to the *global event structure* of a quantum system, there *always exists a local two-valued truth-functional* assignment with respect to a *complete Boolean algebra of projection operators* on the Hilbert space of a quantum system, identified with a *complete Boolean subalgebra* of the global non-Boolean event algebra of a quantum system. More precisely, each *self-adjoint operator* representing an observable has *associated with it* a Boolean subalgebra which is identified with the Boolean algebra of projection operators belonging to its *spectral decomposition*. Hence, given a set of observables of a quantum system, there always exists a complete Boolean algebra of projection operators, viz., a *local Boolean subalgebra* of the global non-Boolean event algebra of a quantum system with respect to which a local two-valued truth-functional assignment is meaningful, if and only if the given observables are *simultaneously measurable*. Consequently, the possibility of local two-valued truth-functional assignments of the global non-Boolean event algebra of a quantum system points to the conclusion that *complete Boolean algebras play the topological role of local Boolean logical frames for contextual true/false value assignments*.

Because of this, we propose a shift in the semantics of quantum events from the set-theoretic level to the sheaf-theoretic, topological level. In particular, the epistemic path proposed in this volume implements the intuitively clear idea of probing the structure of a quantum algebra of events in terms of *localizing Boolean contexts*, admitting an unquestionably operational interpretation. In this regard, the quantum-theoretic localization process we describe involves the following: First, the introduction of a *localization scheme* consisting of (local or partial information) Boolean algebras *completely covering* a quantum algebra of quantum events by means of coordinatizing morphisms. These morphisms from the Boolean localizing reference contexts capture, in essence, separate, *complementary potential relations*, thus providing a structural decomposition or *classification* of a quantum event algebra in the descriptive terms of local Boolean reference contexts. These contexts thus provide potentiality windows for the actualization of concrete events via the evaluation or measurement of the observables contextualized. Second, the coherent realization of the quantum-theoretic localization process demands the satisfaction of *partial compatibility* between *overlapping* local Boolean covers. This guarantees an efficient gluing *procedure* between different coordinatizations of a quantum algebra of events via the measurement of observables. Third, the qualitative content incorporated in local Boolean reference contexts should be

*integrated globally*, such that it respects the requirement of compatibility under both *extension* from the local level to the global level, and *restriction* from the global level to the local level.

### 8.3 Logical Internal Relation: Coherent

#### Local-to-Global Ontological Formation

The above three constituent characteristics capture the essence of the sheaf-theoretic semantics of a quantum event algebra induced by a localization scheme of Boolean reference contexts. Thus, every *local observable (potential fact)* is *internally related*, via the *congruence* and *compatibility* conditions induced by the localization scheme, to a global quantum event structure of actualities. We will now proceed to explain how these relations are expressed in terms of *covering sieves* of quantum event algebras.

The *mutually implicative internal relation* between the local Boolean quantum level and the global, partially Boolean quantum level, is formulated mathematically via a sheaf-theoretic semantics in terms of *covering sieves of a quantum event algebra*. This mereotopological, internal relational structure is the mechanism by which local, intra-contextual logical implication is both restricted and extended globally, inter-contextually. Before defining precisely this concept of covering sieves of a quantum event algebra, it is important to stress the consequences referring to the interpretation of a ‘quantum event continuum’ via its sheaf-theoretic conceptualization. As emphasized earlier, *a quantum event continuum is not given as a completed, already extant set-theoretic structure; rather, it is a mereotopological process of continuous ontological formation that unfolds via the punctuated actualization of potential facts with respect to their local Boolean contexts*. Most significantly, this process of formation is *coherent* in the sense of respecting the rules of transition from the local or partial to the global and conversely. Moreover, the actualization of any potential fact is an *objective augmentation* of the already formed global event structure. The functional dependence inherent in sheaf-theoretic localization thus *relativizes* the presupposed rigid relations between quantum events with respect to *variable* local Boolean contexts *conditioning* the actualization of events. In this way, each *new* actualized event creates a *novelty integrated whole*, and this is due to the fact that its relational integration to the previously established totality is possible only via gluing and compatibility. It is elucidative to quote Alfred North Whitehead in this respect:

The ultimate metaphysical principle is the advance from disjunction to conjunction, creating a novel entity other than the entities given in disjunction. The novel entity is at once the togetherness of the “many” which it finds, and also it is one among the disjunctive “many”

which it leaves; it is a novel entity, disjunctively among the many entities which it synthesizes. The many become one and are increased by one. In their natures, entities are disjunctively “many” in process of passage to conjunctive unity . . . Thus the production of “novel togetherness” is the ultimate notion embodied in the term “conrescence.”<sup>5</sup>

Thus, when Whitehead says that the “many become one and are increased by one” this can be interpreted from a sheaf-theoretic perspective as follows: The many become one by compatible interconnection or *gluing* of their relational information content (encompassing the information of the already actualized events in local partially-compatible contexts), and this process is generative of a novel quantum event; thus, the many are increased by one, because the newly integrated totality now includes this novel event. Together with the compatible local context of this event’s actualization, the novel totality, in turn, imposes further compatibility constraints on new local contexts (specified by the requirements of the conjunctive gluing procedure) for the integration of new observable information, and so on.

The defining conditions characterizing a sheaf incorporate the formal requirements of this process, wherein every local contextual actualization of a potential fact is internally related to a unique global totality of already actualized facts (events). It is necessary to emphasize that the sheaf-theoretic localization of quantum observability is thus conceived as an *ontological process* and not as a mere epistemic restructuring of a fixed, perpetually extant global totality of quantum events. In this sense it provides a conceptual avenue for a further category-theoretic investigation of the already established relevance and suitability of Whitehead’s process philosophy for the interpretation of quantum mechanics.<sup>6</sup> For example, a global quantum event algebra, although hypothesized to exist in a way corresponding to reality, is *not* given as a completed whole initially, but has to be *formed via actualization* of local potential facts with respect to Boolean contexts adding precisely factual content to the global. Furthermore, the *bidirectional form* of consistent internal relation depicted via sheaf-theoretic localization reflects the fact that every potential quantum event, actualized with respect to a local Boolean context, is always logically correlated (by means of an equivalence relation) to any other potential quantum event actualized with respect to any other compatible local Boolean context. *Globally, this means that quantum observables are not determined by their measured values, but by their Boolean observable germs, viz., equivalence classes of compatible Boolean observables forming an inductive limit* . Concomitantly, global quantum events are determined by evaluations of Boolean observable germs.

## 8.4 Functor of Boolean Reference Frames

A *Boolean shaping functor* or functor of *local Boolean coefficients* of a quantum categorical event structure,  $\mathbf{M} : \mathcal{B} \rightarrow \mathcal{L}$ , assigns to Boolean event algebras in  $\mathcal{B}$  (constitutive of a model coordinatizing category) the underlying quantum event algebras from  $\mathcal{L}$ , and likewise assigns to Boolean homomorphisms the underlying quantum algebraic homomorphisms. Throughout the remainder of the discussion, for reasons of simplicity in the notation, the Boolean shaping functor will not appear explicitly in the formulas, although it will be always implicitly assumed.

If  $\mathcal{B}^{op}$  is the opposite category of  $\mathcal{B}$ , then  $\mathbf{Sets}^{\mathcal{B}^{op}}$  denotes the functor category of *presheaves* on Boolean event algebras, with objects all functors  $\mathbf{P} : \mathcal{B}^{op} \rightarrow \mathbf{Sets}$ , and morphisms all natural transformations between such functors. Each object  $\mathbf{P}$  in this category is a contravariant set-valued functor on  $\mathcal{B}$ , called a *presheaf* on  $\mathcal{B}$ . For each Boolean event algebra  $B$  of  $\mathcal{B}$ ,  $\mathbf{P}(B)$  is a set, and for each arrow  $f : C \rightarrow B$ ,  $\mathbf{P}(f) : \mathbf{P}(B) \rightarrow \mathbf{P}(C)$  is a set-function. If  $\mathbf{P}$  is a presheaf on  $\mathcal{B}$  and  $p \in \mathbf{P}(B)$ , the value  $\mathbf{P}(f)(p)$  for an arrow  $f : C \rightarrow B$  in  $\mathcal{B}$  is called the *restriction* of  $x$  along  $f$  and is denoted by  $\mathbf{P}(f)(p) = p \cdot f$ .

The *category of elements* of a presheaf  $\mathbf{P}$ , denoted by  $\int(\mathbf{P}, \mathcal{B})$  has as its objects all pairs  $(B, p)$ , and morphisms  $(B', p') \rightarrow (B, p)$  are those morphisms  $u : B' \rightarrow B$  of  $\mathcal{B}$  for which  $p \cdot u = p'$ .

Projection on the second coordinate of  $\int(\mathbf{P}, \mathcal{B})$ , defines a functor  $\int_{\mathbf{P}} : \int(\mathbf{P}, \mathcal{B}) \rightarrow \mathcal{B}$ .  $\int(\mathbf{P}, \mathcal{B})$  together with the projection functor  $\int_{\mathbf{P}}$  is defined as the *split discrete fibration* induced by  $\mathbf{P}$ , where  $\mathcal{B}$  is the base category of the fibration as in the diagram below.

$$\begin{array}{ccc} \int(\mathbf{P}, \mathcal{B}) & & \\ \downarrow \int_{\mathbf{P}} & & \\ \mathcal{B} & \xrightarrow{\mathbf{P}} & \mathbf{Sets} \end{array}$$

We note that the fibers are categories in which the only arrows are identity arrows. If  $B$  is an object of  $\mathcal{B}$ , the inverse image under  $\int_{\mathbf{P}}$  of  $B$  is simply the set  $\mathbf{P}(B)$ , although its elements are written as pairs so as to form a disjoint union.

The *representation* or *realization* functor of the category of quantum event algebras  $\mathcal{L}$  into the category of presheaves of Boolean event algebras  $\mathbf{Sets}^{\mathcal{B}^{op}}$  is defined by:

$$\mathbf{R} : \mathcal{L} \rightarrow \mathbf{Sets}^{\mathcal{B}^{op}}$$



The *functor of Boolean reference frames* of a quantum event algebra  $L$  in  $\mathbf{L}$  is the *image* of the representation functor  $\mathbf{R}$ , evaluated at  $L$ , into the category of presheaves of Boolean events algebras  $\mathbf{Sets}^{\mathbf{B}^{op}}$ , that is:

$$\mathbf{R}(L) := \mathbf{R}_L : \mathbf{B}^{op} \rightarrow \mathbf{Sets}$$

Notice that the representation functor of  $L$  is completely determined by the action of the functor of Boolean reference frames, for each quantum event algebra  $L$  in  $\mathbf{L}$ , on the objects and arrows of the category of Boolean event algebras  $\mathbf{B}$ , specified as follows: Its action on an object  $B$  in  $\mathbf{B}$  is given by

$$\mathbf{R}(L)(B) := \mathbf{R}_L(B) = \text{Hom}_{\mathbf{L}}(B, L)$$

whereas its action on a morphism  $x : D \rightarrow B$  in  $\mathbf{B}$ , for  $v : B \rightarrow L$  is given by

$$\mathbf{R}(L)(x) : \text{Hom}_{\mathbf{L}}(B, L) \rightarrow \text{Hom}_{\mathbf{L}}(D, L)$$

$$\mathbf{R}(L)(x)(v) = v \circ x$$

It is important to stress that the functor of Boolean reference frames of a quantum event algebra  $L$  in  $\mathbf{L}$ , is a *presheaf*  $\mathbf{R}(L) := \mathbf{R}_L : \mathbf{B}^{op} \rightarrow \mathbf{Sets}$ . Thus, we can legitimately consider the category of elements  $\int(\mathbf{R}_L, \mathbf{B})$ , together with the projection functor  $\int \mathbf{R}_L : \int(\mathbf{R}_L, \mathbf{B}) \rightarrow \mathbf{B}$ , viz., the split discrete fibration induced by the functor of Boolean frames of  $L$ , where  $\mathbf{B}$  is the base category of Boolean reference contexts of the fibration. Hence, the functor of Boolean frames of a quantum event algebra induces a *uniform and homologous fibered representation of quantum events in terms of Boolean reference frames* for the measurement of observables.

Thus, in mathematical terms, the quantum localization process in the descriptive terms of Boolean reference frames is understood by means of an *action* of the category of (local) Boolean event algebras on a set-theoretic global structure of quantum events. A global quantum event algebra is then *partitioned* into sorts parameterized by the *Boolean frames* of the base category of local Boolean contexts. In this way, the functioning of a Boolean localization scheme on a global structure of quantum events is represented by means of a *fibered construct*, understood geometrically, as a locally variable set over the base category of local Boolean reference contexts. The notion of functional dependence incorporated in this action forces *the partially ordered structure of quantum events to fiber over the base category of local Boolean reference contexts*.

It is important to emphasize that the functor of Boolean reference frames of a quantum event algebra  $L$ , viz., *the network of relationships* that  $L$  has with all Boolean contexts  $B$ , formalized as a presheaf, incorporates the physical requirement of *uniformity* among events. By this we mean that for any two

quantum events observed over the same Boolean domain of measurement, the structure of all Boolean logical reference frames that relate to the first cannot be distinguished in any possible way from the structure of Boolean frames relating to the second. According to the principle of uniformity, in other words, all the localized quantum events within any particular Boolean reference context should be *uniformly equivalent* to each other. The compatibility of the Boolean localization process of a quantum event algebra with the requirement of uniformity entails that the partial order of relations in a global structure of quantum events is induced by lifting an appropriate family of morphisms from the base category of local Boolean reference contexts to the fibers. Equivalently, *the partial order of relations among quantum events is induced by corresponding relations among their localizing Boolean reference contexts.*

## 8.5 Functorial Boolean-Quantum

### Internal Relation

The network of relationships defined by a quantum event algebra with Boolean logical frames, formalized categorically by the notion of a presheaf functor of Boolean frames inducing a Boolean localization scheme, is the semantic information carrier of a *mutually implicative internal relation* between the local Boolean level and the global quantum level. This *dipolar* mutually implicative internal relation is formulated in the category theoretic syntax in terms of a pair of adjoint functors between the category of presheaves of Boolean event algebras and the category of quantum event algebras, thus forming a categorical *adjunction*. This categorical adjunction formalizes the process-theoretic operation of the category of *transmutation* by relating internally and bidirectionally the local Boolean and global quantum levels of the event structure. More precisely, we formulate the following proposition:<sup>7</sup>

*There exists a pair of adjoint functors  $\mathbf{L} \dashv \mathbf{R}$  as follows:*

$$\mathbf{L} : \mathbf{Sets}^{\mathbf{B}^{op}} \rightleftarrows \mathbf{L} : \mathbf{R}$$

The Boolean frames-quantum adjunction consists of the functors  $\mathbf{L}$  and  $\mathbf{R}$ , called left and right adjoints with respect to each other respectively, as well as the natural bijection:

$$\mathbf{Nat}(\mathbf{P}, \mathbf{R}(L)) \cong \mathbf{Hom}_{\mathbf{L}}(\mathbf{LP}, L)$$

The above bijective correspondence, interpreted functorially, says that the *Boolean realization functor* of a quantum categorical event structure  $\mathbf{L}$ , realized for each quantum event algebra  $L$  in  $\mathbf{L}$  by its *functor of Boolean frames*, viz., by

$$\mathbf{R}(L) : B \mapsto \text{Hom}_L(B, L)$$

has a *left adjoint* functor  $\mathbf{L} : \text{Sets}^{\mathcal{B}^{op}} \rightarrow \mathcal{L}$ , which is defined for each presheaf of Boolean algebras  $\mathbf{P}$  in  $\text{Sets}^{\mathcal{B}^{op}}$  as the colimit (inductive limit):

$$\mathbf{L}(\mathbf{P}) = \text{Colim}\{(\mathbf{P}, \mathcal{B}) \rightarrow \mathcal{B} \rightarrow \mathcal{L}\}$$

Thus, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{B} & & \\ \downarrow \mathbf{y} & \searrow \mathbf{M} & \\ \text{Sets}^{\mathcal{B}^{op}} & \xrightarrow{\quad \mathbf{L} \quad} & \mathcal{L} \end{array}$$

The physical meaning of the adjunction between presheaves of Boolean logical frames and quantum event algebras is made transparent if we consider that the mutually inverse pair of internally related adjoint functors formalizes the *mutually implicative processes of encoding and decoding information* relevant to the structural form of their domain and codomain categories. If we think of  $\text{Sets}^{\mathcal{B}^{op}}$  as the categorical universe of variable local Boolean frames modeled in *Sets*, and of  $\mathcal{L}$  as the categorical universe of quantum event structures, then the functor  $\mathbf{L} : \text{Sets}^{\mathcal{B}^{op}} \rightarrow \mathcal{L}$  signifies a *translational code* of information from the level of locally Boolean events to the level of global quantum events; and likewise, the Boolean realization functor  $\mathbf{R} : \mathcal{L} \rightarrow \text{Sets}^{\mathcal{B}^{op}}$  signifies a translational code in the inverse direction.

In general, the content of the information cannot remain completely invariant with respect to translation from one categorical universe to another and conversely. However, there remain two alternatives for a variable set over local Boolean algebras  $\mathbf{P}$  to exchange information with a quantum algebra  $L$  : *either* the content of information is exchanged in *quantum terms* with the inductive limit in the category of elements of  $\mathbf{P}$  translating, represented as the quantum morphism  $\mathbf{LP} \rightarrow L$ ; *or* the content of information is exchanged in *Boolean terms* with the functor of Boolean frames of  $L$  translating, represented correspondingly as the natural transformation  $\mathbf{P} \rightarrow \mathbf{R}(L)$ .

In the first case, from the perspective of  $L$ , information is being received in quantum terms while in the second, from the perspective of  $\mathbf{P}$ , information is being sent in Boolean terms. The natural bijection then corresponds to the assertion that these *two distinct ways of communicating are equivalent*. Thus, the

fact that these two functors are adjoint expresses a relation of variation, regulated by *two poles*, with respect to the meaning of the information related to observation. We claim that the totality of the content of information included in quantum observable structures remains *invariant under Boolean encodings*, corresponding to local Boolean logical frames, *if and only if* the adjunctive correspondence can be appropriately restricted to an *equivalence* of the functorially correlated categories. This task can be accomplished by defining an appropriate Grothendieck topology on the base category of Boolean frames, which essentially paves the way for the understanding of a quantum event algebra as a *sheaf* of local Boolean algebras over an appropriately specified covering system of the former. This categorical equivalence is interpreted physically in terms of a *transmuted information invariance property* between the Boolean and the quantum species of structure, referring to the *bidirectional internal relation* between variable local Boolean algebras and global quantum ones.

For any presheaf  $\mathbf{P}$  in  $\mathbf{Sets}^{B^{op}}$ , the *unit* natural transformation of the Boolean frames-quantum adjunction is defined as follows:

$$\delta_{\mathbf{P}} : \mathbf{P} \rightarrow \mathbf{RLP}$$

Likewise, for each quantum event algebra  $L$  in  $\mathbf{L}$  the *counit* natural transformation is defined as follows:

$$\varepsilon_L : \mathbf{LR}(L) \rightarrow L$$

The representation of a quantum event algebra  $L$  in  $\mathbf{L}$ , in terms of the functor of Boolean frames  $\mathbf{R}(L)$  of  $L$ , is *full and faithful*, if and only if the counit is a *quantum algebraic isomorphism*. In turn, the counit is a quantum algebraic isomorphism if and only if the right adjoint functor is full and faithful. In the latter case we characterize the Boolean shaping functor  $\mathbf{M} : \mathbf{B} \rightarrow \mathbf{L}$  as a *proper* or *dense shaping functor*. It can be shown that the Boolean realization functor is full and faithful if it corresponds to a *functor of Boolean localizations* of  $L$  for an appropriate *Grothendieck topology*.<sup>8</sup> Thus, the counit is an isomorphism if it is restricted to some *Boolean localization system* of  $L$ .

From the above, we deduce that the representation of a quantum event algebra  $L$  in  $\mathbf{L}$ , in terms of the functor of Boolean frames  $\mathbf{R}(L)$  of  $L$ , is full and faithful, if the Boolean frames-quantum adjunction is *restricted* to a Boolean localization system of  $L$ . As a corollary we obtain that a quantum categorical event structure  $\mathbf{L}$  is a *reflection* of the topos of presheaves  $\mathbf{Sets}^{B^{op}}$ .

## 8.6 Quantum Localization:

### Sheaf of Boolean Reference Frames

According to the defining requirements of a localization scheme, the established fibred representation of quantum events, realized for each quantum event algebra by its functor of Boolean frames, should be also coherent under extension from the local to the global level and conversely. Thus, we should define an appropriate Grothendieck topology  $\mathbf{J}$  on the base category of Boolean contexts  $\mathbf{B}$ , such that: [i] The Boolean reference frames acquire the semantics of *local frames* with respect to that Grothendieck topology  $\mathbf{J}$  on  $\mathbf{B}$ , and [ii] The functor of Boolean frames for a quantum event algebra  $L$  in  $\mathbf{L}$  becomes a *sheaf on the site*  $(\mathbf{B}, \mathbf{J})$  for that  $\mathbf{J}$ . For this purpose, we use the machinery of covering sieves of Boolean frames, defined as follows:

A  $B$ -sieve  $S$  on a Boolean reference context  $B$  in  $\mathbf{B}$  is called a *covering sieve* of  $B$ , if all the arrows  $s : C \rightarrow B$  belonging to the sieve  $S$ , taken together, form an *epimorphic family* in  $\mathbf{L}$ . This requirement may be, equivalently, expressed in terms of a map:

$$G_S : \coprod_{(s:C \rightarrow B) \in S} C \rightarrow B$$

being an *epimorphism* in  $\mathbf{L}$ . It can be proved that the specification of covering sieves on Boolean contexts  $B$  in  $\mathbf{B}$ , in terms of epimorphic families of arrows in  $\mathbf{L}$ , *does indeed define a categorical topological covering system (Grothendieck topology)  $\mathbf{J}$  on  $\mathbf{B}$* .<sup>9</sup> Thus, the function  $\mathbf{J}$ , which assigns to each Boolean reference context  $B$  in  $\mathbf{B}$ , a collection  $\mathbf{J}(B)$  of covering  $B$ -sieves, being epimorphic families of arrows in  $\mathbf{L}$ , constitutes a Grothendieck topology  $\mathbf{J}$  on  $\mathbf{B}$ . Then, we can show that the *functor of Boolean reference frames  $\mathbf{R}_L$*  is a *sheaf for the Grothendieck topology  $\mathbf{J}$* , defined by covering sieves of epimorphic families on the category of Boolean reference contexts, as above.<sup>10</sup> Hence, the Boolean reference frames  $B \rightarrow L$  acquire the *semantics of local Boolean frames* with respect to the Grothendieck topology  $\mathbf{J}$ , defined in terms of covering sieves of epimorphic families of arrows in  $\mathbf{L}$ . Moreover, since the functor of Boolean frames for a quantum event algebra  $L$  becomes a *sheaf* restricted on the site  $(\mathbf{B}, \mathbf{J})$  for that  $\mathbf{J}$ , the split discrete fibration  $\mathbf{J}\mathbf{R}_L : \mathbf{J}(\mathbf{R}_L, \mathbf{B}) \rightarrow \mathbf{B}$ , induced by the *sheaf of Boolean frames* of  $L$  on the site  $(\mathbf{B}, \mathbf{J})$ , forces a uniform, homologous, and also coherent *fibered representation of quantum events in terms of local Boolean reference frames*.

More concretely, the definition of the Grothendieck topology  $\mathbf{J}$  on the base category of Boolean reference contexts  $\mathbf{B}$ , induces the functionality of a quantum localization scheme, formulated in terms of covering sieves on a quantum event algebra  $L$  as follows: We say that a sieve on a quantum event algebra  $L$  defines a covering sieve by Boolean reference frames, such that the domains of the frames belonging to the sieve are Boolean contexts in  $\mathbf{B}$ , if all these morphisms in the sieve collectively define an *epimorphism* (that is, collectively *cover* it completely):

$$T : \coprod_{(E \in [\mathbf{B}]_0, \not\vdash_E E \rightarrow L)} E \rightarrow L$$

where  $[\mathbf{B}]_0$  denotes the set of Boolean reference contexts of the base category  $\mathbf{B}$ . From the physical point of view, *covering sieves of a quantum event algebra by local Boolean reference frames are interpreted as Boolean localization systems of a quantum event algebra.*<sup>11</sup> It is instructive to think of covering sieves of a quantum event algebra  $L$  as *subfunctors* of the functor of Boolean frames of  $L$ , viz., of the functor  $\mathbf{R}(L) := \mathbf{R}_L : \mathbf{B}^{op} \rightarrow \mathbf{Sets}$ . These subfunctors of  $\mathbf{R}(L)$ , that is, the covering sieves of  $L$ , are *partially ordered* by inclusion. The partial order relation among covering sieves of a quantum event algebra  $L$  is interpreted as an operation of *coarse-graining* among Boolean localization systems.

## 8.7 Boolean Localization Systems and

### Boolean Germs

A Boolean localization system forces a transition in the semantics of quantum events from the set-theoretic to the topological sheaf-theoretic level. This is due to the following proposition:<sup>12</sup>

*A covering sieve of a quantum event algebra  $L$  by local Boolean reference frames, that is, a Boolean localization system of  $L$ , induces an isomorphism:*

$$L \cong \mathbf{Colim}\{\mathcal{J}(\mathbf{R}_L, \mathbf{B}) \rightarrow \mathbf{B}\}$$

where,  $\mathbf{R}_L$  denotes a sheaf of Boolean frames of  $L$  on the site  $(\mathbf{B}, \mathbf{J})$ , for  $\mathbf{J}$  the Grothendieck topology of epimorphic families, and  $\mathcal{J}(\mathbf{R}_L, \mathbf{B})$  is the associated fibration induced by the category of Boolean frames of  $L$ .

Thus, a quantum event algebra  $L$  is represented *isomorphically* by the *inductive limit taken in the category of local Boolean reference frames of the sheaf  $\mathbf{R}_L$*  with respect to a *Boolean localization system*. Let us now explain the meaning of this *inductive limit* (colimit) construction. For this purpose we consider the set:

$$\mathbf{Y}(\mathbf{R}_L) = \{(\psi_B, q) / (\psi_B : B \rightarrow L), q \in B\}$$

Note that the morphisms  $\psi_B : B \rightarrow L$  denote Boolean reference frames of  $L$ , encoded as elements in the category of Boolean frames of  $L$ , viz.,  $\mathcal{J}(\mathbf{R}_L, \mathbf{B})$ . We notice that if there exists a Boolean morphism  $u : B'$

$\rightarrow B$  such that:  $u(q') = q$  and  $\psi_B \cdot u = \psi_{B'}$ , by restriction along  $u : B' \rightarrow B$ , then we may define a *transitive* and *reflexive* relation  $\#$  on the set  $\mathbf{Y}(\mathbf{R}_L)$ . Of course the inverse also holds true. Then we obtain the following:

$$(\psi_B \cdot u, q) \mathfrak{R}(\psi_B, u(q'))$$

for any  $u : B' \rightarrow B$  in the category  $\mathbf{B}$ . The next step is to make this relation also *symmetric* by postulating that for  $\zeta, \eta$  in  $\mathbf{Y}(\mathbf{R}_L)$ , where  $\zeta, \eta$  denote pairs in the above set, we have:

$$\zeta : \eta$$

if and only if  $\zeta \# \eta$  or  $\eta \# \zeta$ . Finally by considering a sequence  $\xi_1, \xi_2, \dots, \xi_k$  of elements of the set  $\mathbf{Y}(\mathbf{R}_L)$  and also  $\zeta, \eta$  such that there exists a path of Boolean transition arrows:

$$\zeta : \xi_1 : \xi_2 : \dots : \xi_{k-1} : \xi_k : \eta$$

We may define an equivalence relation on the set  $\mathbf{Y}(\mathbf{R}_L)$  as follows:

$$\zeta \triangleright \triangleleft \eta := \zeta : \xi_1 : \xi_2 : \dots : \xi_{k-1} : \xi_k : \eta$$

Then for each  $\zeta = (\psi_B, q) \in \mathbf{Y}(\mathbf{R}_L)$  we define the Boolean germ (equivalence class) at  $\zeta$  as follows:

$$Q_\zeta = \{t \in \mathbf{Y}(\mathbf{R}_L) : \zeta \triangleright \triangleleft t\}$$

We finally define:

$$\mathbf{Y}(\mathbf{R}_L) / \triangleright \triangleleft := \{Q_\zeta : \zeta = (\psi_B, q) \in \mathbf{Y}(\mathbf{R}_L)\}$$

and use the notation  $Q_\zeta = P(\psi_B, q)P$ , where  $P(\psi_B, q)P$  denotes the *Boolean germ* at  $\zeta = (\psi_B, q)$ . The *quotient set*  $\mathbf{Y}(\mathbf{R}_L) / \triangleright \triangleleft$  defines the *inductive limit* in the category of local Boolean reference frames of the sheaf  $\mathbf{R}_L$ , that is,

$$\mathbf{Y}(\mathbf{R}_L) / \triangleright \triangleleft = \mathbf{Colim} \{ \int(\mathbf{R}_L, \mathbf{B}) \rightarrow \mathbf{B} \}$$

by noticing that it is naturally endowed with a quantum event algebra structure as follows:

[1] The orthocomplementation is defined as:

$$Q_\zeta^* = P(\psi_B, q)P = P(\psi_B, q^*)P.$$

[2] The unit element is defined as:  $\mathbf{1} = P(\psi_B, 1)P$ .

[3] The partial order structure on the quotient set  $\mathbf{Y}(\mathbf{R}_L)/\triangleright \triangleleft$  is defined as:  $P(\psi_B, q)P\# P(\psi_C, r)P$  if and only if  $d_1 \# d_2$  where we have made the following identifications:  $P(\psi_B, q)P = P(\psi_D, d_1)P$  and  $P(\psi_C, r)P = P(\psi_D, d_2)P$ , with  $d_1, d_2 \in D$  according to the pullback diagram of event algebras:

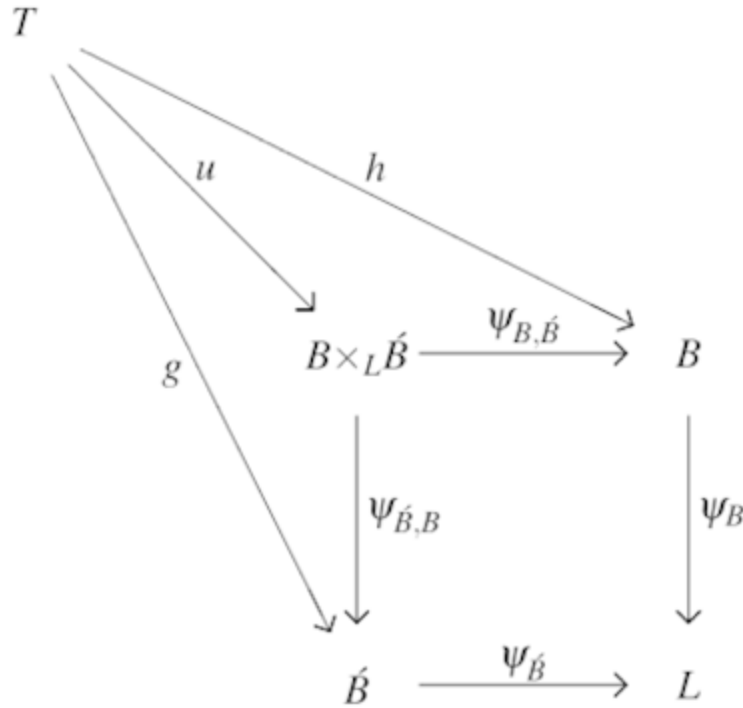
$$\begin{array}{ccc}
 D & \xrightarrow{\beta} & B \\
 \downarrow \gamma & & \downarrow \\
 C & \xrightarrow{\quad} & L
 \end{array}$$

such that  $\beta(d_1) = q$ ,  $\gamma(d_2) = r$ .

## 8.8 Gluing Isomorphism of Local Boolean Reference Frames

In order to understand the above pullback diagram we have to remember that  $\mathbf{R}_L$  is a *sheaf of Boolean frames* of  $L$  on the site  $(\mathbf{B}, \mathbf{J})$ , for  $\mathbf{J}$  the Grothendieck topology of *epimorphic families*. The fact that  $\mathbf{R}_L$  is a sheaf of Boolean frames of  $L$  means the following: Let us consider two Boolean reference frames  $\psi_B : B \rightarrow L$  and  $\psi_{B'} : B' \rightarrow L$  of  $L$ . Then the *pullback (categorical overlap)* of the Boolean reference frames  $\psi_B$  and  $\psi_{B'}$ , with common codomain a quantum event algebra  $L$ , consists of the event algebra  $B \times_L B'$  and two arrows  $\psi_{BB'}$  and  $\psi_{B'B}$ , called *projections*, as shown in the following diagram:





The *square commutes* and for any event algebra  $T$  and arrows  $h$  and  $g$  that make the outer square commute, there is a unique  $u : T \rightarrow B \times_L B'$  that makes the whole diagram commute. Now, if  $\psi_B$  and  $\psi_{B'}$  are injective morphisms, then their pullback is isomorphic with their intersection  $B \cap B'$ .

The fact that the functor  $\mathbf{R}_L$  of Boolean frames of  $L$  is a sheaf, means that there exists a *gluing isomorphism* of Boolean reference frames on their overlap, defined as follows:

$$\Omega_{B,B'} : \psi_{B'B}(B \times_L B') \rightarrow \psi_{BB'}(B \times_L B')$$

$$\Omega_{B,B'} = \psi_{BB'} \circ \psi_{B'B}^{-1}$$

The gluing isomorphism  $\Omega_{B,B'}$  means that the Boolean reference frames of  $L$ , that is,  $\psi_{B'B}(B \times_L B')$  and  $\psi_{BB'}(B \times_L B')$ , cover the same part of a quantum event algebra  $L$  in a compatible way. Furthermore, an immediate consequence of the previous definition is the satisfaction of the following *cocycle conditions*:

$$\begin{aligned}
 \Omega_{B,B} &= id_B \\
 \Omega_{B,B'} \circ \Omega_{B',B''} &= \Omega_{B,B''} \quad \text{if } B \times_L B' \times_L B'' \neq 0
 \end{aligned}$$

$$\Omega_{B,B'} = \Omega_{B',B}^{-1} \quad \text{if } B \times_L B' \neq 0$$

We conclude that the gluing isomorphism between two Boolean reference frames  $\Omega_{B'B'}$  in the sheaf of Boolean frames  $\mathbf{R}_L$  assures that  $\psi_{B'B}(B \times_L B')$  and  $\psi_{BB'}(B \times_L B')$  are being *glued and thus cover* the same part of a quantum event algebra  $L$  in a compatible way.

Consequently, we state the following conclusion: *A quantum event algebra  $L$  can be represented isomorphically by the inductive limit taken in the category of elements of the sheaf functor of Boolean reference frames of  $L$ .* Hence, a quantum event algebra can be represented by means of a *sheaf of Boolean germs*, with respect to the topology  $\mathbf{J}$ . Globally, this means that quantum observables are *not* determined by their measured values, but they *are* determined by their *Boolean observable germs*, viz., equivalence classes of partially compatible Boolean observables. Concomitantly, quantum events are specified by *evaluations* of their corresponding Boolean germs with respect to Boolean localization systems. Thus, there exists a uniform, homologous and coherent fibered representation of quantum events with respect to local Boolean reference frames forming Boolean localization systems of a quantum event algebra.

We conclude that every local observable (potential fact) is *internally related*, via the congruence and compatibility relations induced by the correlate Boolean localization system, to a global quantum event structure of actualities. Conversely, a global quantum event continuum is *not* given as a completed a priori set-theoretic structure; rather, it is constituted, in a *continuous* topological process of ontological formation, by *actualizations of potential facts* with respect to local Boolean contexts. These actualizations are expressed categorically by *evaluations of Boolean germs*. Thus, given the functional role of Boolean germs as internally constitutive of becoming actual occasions, this sheaf-theoretic process of continuous formation is fully coherent in the sense of respecting the rules of *extension* from the local to the global, and *restriction* of the local by the global.

## Notes

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10. See note 9.

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